

Methodology and Architectures

Common-recipe protocol, in-house families, external references, and negative candidates

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Verified common scope	43 models; five opened datasets; 96,320 exact jobs
Primary outcome	Equal-dataset macro balanced accuracy
Observed leader	cardinal_fbc_compactdyn_scale025_extended (73.9900%)
Experimental decisions	Gauge Gate 1 failed; CHSD promotion stopped
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Methodology and architecture catalogue

Scope of this document

This is the publication-facing methods summary for the clean project. The source of truth for executable common-grid identities is `benchmark.full_grid.COMMON_ARCHITECTURES` plus `benchmark.baselines.make_model`. The broader ownership/provenance crosswalk is `docs/INHOUSE_MODEL_METHODS.md` and the 70-entry registry is `src/benchmark/model_registry.py`.

The token `eeg-mi-cache-v2` below is an immutable cache-schema identifier from the completed protocol. It is not the active package or distribution name; those are `benchmark` and `benchmark`.

"In-house" means that this repository owns the implementation or the project-specific derivative/procedure. It does not establish legal ownership, patentability, literature novelty, clinical validity, or state of the art. A compact, extended, wide, scale, pooling, dropout, loss, or initialization variant is normally a configuration or ablation, not a new architecture.

Verified common-recipe experiment

The completed common grid compared 43 raw-trial configurations on five previously opened development datasets. Its immutable product was:

`43 models x 448 dataset/participant/fold units x 5 seeds = 96,320 jobs`

The participant/fold dimensions were:

Dataset	Participants	Folds per participant	Classes
Local Exp4	8	1 fixed chronological fold	2
BNCI2014-001	9	1 official-session fold	4
BNCI2014-004	9	1 official-session fold	2
Cho2017	52	5 acquisition-order block folds	2
PhysioNet MI development	54	3 imagery-run folds	2

The seeds were 7, 17, 27, 37, and 47. Every common configuration used the same data split, scaling, augmentation, optimizer, stopping, reset/refit, and test prediction contract. Consequently, the table answers a shared-recipe architecture/configuration question. It is not an author-recipe reproduction for every external model.

Preprocessing

Public-dataset trials use the `eeg-mi-cache-v2` harmonized profile: 4-40 Hz, 0.5-3.0 s half-open epochs, 128 Hz, 320 samples, epoch demeaning, and symmetric common-average reference, except that BNCI2014-004 retains its three supplied bipolar derivations. The common montage profile uses 21 declared channels where available. Local Exp4 uses 4-40 Hz, 0.0-2.0 s half-open, 128 Hz/256 samples, epoch demeaning, and common-average reference over 15 point electrodes. Details and rights boundaries are in `docs/DATA_ACCESS.md`.

Channel standardization is source-only. Selection training rows determine the selection scaler. Train plus validation rows determine a new final-refit scaler. Test rows are excluded from both.

Selection, reset/refit, and prediction

Within every participant/fold/seed job:

- 1 install the seed before model construction;
- 2 optimize only on training rows and select the epoch count by validation cross-entropy;
- 3 reconstruct the exact seeded initialization;
- 4 refit from that initialization on train plus validation for the selected number of epochs; and
- 5 predict the held-out test rows once.

The score-blind production package stores row identities and normalized class probabilities but not labels or performance. The analyzer joins labels only after an exact 96,320-job audit.

Shared optimization recipe

Setting	Frozen value
Maximum epochs	200
Batch size	64
Optimizer	AdamW
Learning rate	0.0008
Weight decay	0.0005
Scheduler	Cosine annealing over the 200-epoch horizon
Early-stopping patience	35
Minimum validation-loss improvement	0.0001
Label smoothing	0.05
Gradient-norm clip	5.0
Segment reconstruction	8 segments, batch-level probability 0.5
Temporal shift	Uniform integer shift up to +/-8 samples, zero padded
Additive Gaussian noise	Standard deviation 0.01 after channel scaling
Reflection augmentation/loss	Disabled in the common recipe

Torch deterministic algorithms were required and `CUBLAS_WORKSPACE_CONFIG=: 4096:8` was bound into formal execution. This is a deterministic software contract for the recorded environment, not a promise that different hardware or library versions will produce byte-identical floating-point results.

Analysis

The primary outcome is balanced accuracy. Folds are concatenated within each participant and seed; seeds are averaged within participant; participants are weighted equally within each dataset; and the five dataset means are weighted equally. Trials, folds, and seeds are not treated as independent participants.

Secondary outcomes are accuracy, chance-normalized balanced accuracy, macro F1, Cohen's kappa, one-vs-rest macro AUROC, negative log likelihood, multiclass Brier score, and 15-bin expected calibration error. Parameter count, selected epochs, fit/prediction time, and peak CUDA allocation are engineering outcomes. Trial-micro results remain separate because they weight datasets by trial count and pool repeated seed predictions.

The fixed-suite descriptive intervals use 100,000 deterministic participant-cluster bootstrap resamples. Seeds are derived reproducibly from the frozen base seed 20260729 and the comparison's frozen model index; the reported leader-versus-TCFormer interval uses seed 20530729. Because the leader was selected from 43 outcomes, its contrast with TCFormer is descriptive and not a selection-adjusted confirmatory confidence interval.

Common-grid in-house architecture families

SCOPE

ScopeNet combines ordered physical-frequency Sinc filters with continuous spherical-polynomial scalp fields and multiresolution energy/dynamics moments. An axial frequency/source/feature mixer is a zero-started residual over the direct energy floor. FreeScopeNet replaces the coordinate field with an indexed spatial matrix and is the continuity control. Common keys: `scope`, `free_scope`.

The control substantially outperformed SCOPE in the verified grid, so these data do not demonstrate a benefit from SCOPE's coordinate-continuous field.

CardinalField

CardinalFieldNet maps coefficients at 21 fixed standard-1005 inducing sites through a regularized spherical Gaussian cardinal basis, then applies 16 ordered Sinc bands, 32 latent sources, four log-variance views, and a classifier. FreeCardinalFieldNet uses fixed-order indexed channel weights. Common keys: `cardinal`, `free_cardinal`.

The interpolation is applied to spatial weights, not EEG trials or labels. The free control exceeded the cardinal version in the verified grid; this is not evidence of a coordinate-field advantage.

CardinalDynamics

CardinalDynamicsNet learns a continuous spatiotemporal cardinal tensor and decodes latent-source log energy plus signed temporal-dynamics summaries. The six common configurations are:

- `cardinal_dynamics`: default free-FIR, 21-anchor form;
- `cardinal_dynamics_compact`: fewer temporal/source/dynamics channels;
- `cardinal_dynamics_extended`: default form with a 31-anchor union atlas;
- `cardinal_dynamics_sinc`: ordered-Sinc compact form;
- `cardinal_dynamics_sinc_residual`: Sinc form plus a gated low-rank multiscale temporal residual; and
- `cardinal_dynamics_sinc_extended`: ordered-Sinc form with 31 anchors.

The Sinc-extended configuration was the observed Local Exp4 leader and ranked third overall under the common recipe. The extended atlas is an ablation; the result does not prove montage invariance or that all 31 inducing degrees of freedom are identifiable.

CardinalFBC

CardinalFBCNet is an in-house derivative of external FBCNet. It preserves FBCNet's spectral filter bank, normalization, activation, temporal log-variance views, and constrained classifier while replacing the grouped indexed spatial convolution with a regularized cardinal-RBF weight field. A fresh indexed layer can be projected into that field to preserve the initial mapping.

Continuation variants keep the CardinalFBC floor and one constrained head; new head columns start at zero:

- base: `cardinal_fbc`, `cardinal_fbc_extended`;
- shrinkage/Fisher-z correlation features: `cardinal_fbc_corr` and its `extended` form;
- compact energy/dynamics continuation: `cardinal_fbc_compactdyn` and its `extended` form;
- fixed continuation scales 0.10 and 0.25, each with 21- and 31-anchor forms;
- learned zero-DC Gabor micro-dynamics: `cardinal_fbc_micro` and its `extended` form; and
- ordered-Sinc physical dynamics: `cardinal_fbc_physical` and its `extended` form.

`cardinal_fbc_compactdyn_scale025_extended` was the observed overall common-grid leader at 73.9900% equal-dataset balanced accuracy. This is an outcome-selected configuration and an FBCNet derivative, not evidence of a wholly independent architecture, confirmed superiority, or global SOTA.

CardinalFBMS

CardinalFBMSNet is the analogous in-house derivative of external FBMSNet. It preserves the public spectral bank, mixed-scale temporal block, normalization, temporal views, and constrained classifier while replacing the indexed spatial convolution with a cardinal field. Common keys: `cardinal_fbms` and `cardinal_fbms_extended` (21 versus 31 anchors).

Do not transfer evidence from the separately trained pretrained-CardinalFBMS procedure to these from-scratch common rows.

CardinalMixedTemporal

CardinalMixedTemporalNet reuses FBMSNet's spectral and mixed-temporal blocks, then uses a full-rank cardinal spatial field and four-segment log-variance decoder. `cardinal_mix_drop` differs from `cardinal_mix` only by decoder dropout 0.25. These are FBMSNet-component derivatives and performed below the leading common configurations.

In-house families outside the common recipe

These families must not be appended to the common leaderboard. Their paired views, SPD features, transfer checkpoints, routing, dataset-specific signals, or author-specific training rules define different scientific questions.

Family	Mechanism and identity	Evidence boundary
GeoAdaptNet / GeoAdaptNet-FB / FBSP	Fixed-band SPD tangent anchor plus gated deep residual, or learned-Sinc filter-bank covariance models with optional rank-8 BiMap	Separate binary geometric/deployment protocol. Residual was largely inert and the family did not establish classifier novelty or superiority.

Family	Mechanism and identity	Evidence boundary
CAMEO	Convex log-Euclidean tangent anchor plus raw energy/dynamics experts on original/reflected views; validation selects subtraction and mixture	Binary procedure; routing is inseparable from the reported result. Historical and current-local protocols cannot be pooled.
HemiParity	Even/odd raw and tangent representations with invariant-context odd readout and reliability gate	Binary reflection-equivariant procedure; not eligible for the four-class task.
PARITY-Fuse	Deterministic paired raw/tangent encoders, invariant odd-latent fusion, and zero-start residual over a convex tangent anchor	Binary procedure; a related but distinct parity family from HemiParity.
HemiQ-FieldNet	Exact reflection-odd cross-spectral field for the three supplied BNCI2014-004 bipolar signals	Dataset-specific. Historical five-subject confirmation was effectively tied with ShallowConvNet; no superiority claim.
CardinalSplineDualView	One CardinalFBC backbone evaluated on native and fixed spherical-spline-transported views, mixed by a detached-disagreement gate	Cho/PhysioNet transfer prototype; no promoted aggregate result or production writer.
OrbitTransportNet	Converts arbitrary paired expert logits into an exactly odd transported score using invariant context; fuses energy, dynamics, and tangent anchor	ORBIT-v1 is the distinct architecture; v2-v5 are configurations/ablations. The local v3 procedure is separate from common training.

Native-transfer procedures

Pretrained/scratch CardinalFBC and CardinalFBMS entries are training procedures, not additional architectures. They use native-montage target caches and, where applicable, a byte-pinned source checkpoint. The target cohorts are Cho2017 S16-S52 and PhysioNet S1-S54; source-only scaling, validation-only epoch selection, initialization reset, train+validation refit, and one-shot outer-test prediction remain mandatory.

The five in-house registry conditions are `pretrained_cardinal_fbc`, `scratch_cardinal_fbc`, `pretrained_cardinal_fbms`, `scratch_cardinal_fbms_canonical_seeded`, and `scratch_cardinal_fbms_native_projected`. The last two distinguish a fresh canonical coordinate-field initialization from projection of a fresh native indexed layer; neither is a new architecture family.

The CardinalFBC transfer family has only limited predecessor coverage under its own protocol. The five-condition CardinalFBMS transfer grid is a separate, audited development artifact; it is not a common-grid substitute and provides no sealed confirmation or clinical evidence.

Negative experimental families

CHSD

The unregistered CHSD family represents native-montage complex coherency as a Hermitian positive-definite surface, uses matrix-log states and explicit time/frequency/mixed differences, and tests direct, residual, joint, and conditioned decoders. Matrix logarithms, complex connectivity, filter banks, coordinate projection, and attention are not individually claimed as novel.

Its executable scratch keys are `chsdnet`, `chsdnet_direct`, `chsdnet_hybrid`, `chsdnet_joint`, and conditioned bounds `chsdnet_conditioned_005`, `chsdnet_conditioned_010`, and `chsdnet_conditioned_020`. They are intentionally unregistered; code existence is not a completed result.

The bounded 0.05 conditioned candidate reached 73.543% equal-dataset balanced accuracy in its separate 115-participant robustness amendment, below the 74.087% CardinalFBC-micro-extended reference, and failed its frozen dataset-breadth and participant-win-rate clauses. The decision was `stop_chsd_promotion`. Other CHSD variants remain code/screen variants, not promoted models.

GaugeQuotientCrossMomentNet

Gauge combines a preserved CardinalFBC path with learned sources, band/window/lag cross moments, mask-specific coordinate quotient features, antisymmetric lag wedges, and a zero-start continuation. The defensible candidate claim was the complete neural composition, not invention of lagged covariance, the exterior product, time asymmetry, average-reference projection, causality, or global invariance. Close equation-level prior art is documented in `docs/GAUGE_QUOTIENT_NOVELTY_PLAN.md`.

Its prespecified 17-subject Gate 1 produced 75.0704% equal-dataset balanced accuracy versus 74.3988% for CardinalFBC-micro-extended, a +0.6716-point margin, and 11/17 subject wins. Nevertheless, dataset deltas were nonnegative on only two of five datasets; the frozen rule required at least three. Gate 1 therefore failed and the persisted decision was `kill_candidate_before_disjoint_gate`. Gate 2, tuning, renamed retries, promotion, and SOTA language are forbidden. Gauge is an informative separate negative result and is not one of the 43 common-grid models.

External common-grid models

The 15 external common configurations are EEGNet, ShallowFBCSPNet, Deep4Net, EEG-Conformer (default and compact), ATCNet (default and aggressive pooling), FBCNet, EEG-TCNet, FBMSNet, CTNet (default and compact), EEGSym (default and wide), and TCFormer. These names, their architectures, and their original papers belong to their respective authors. Local compatibility corrections, shape variants, and use under this project's shared recipe do not change that ownership.

TCFormer uses the compact pinned upstream source at commit `74c89b7ab8c64e4eb51e0f748dd87dd4c94e68c5`; all other public implementations must be traced through the locked software and cited to their original method papers in the manuscript.

Novelty boundary

The completed benchmark can establish reproducible behavior of implemented configurations under its protocol. It cannot by itself establish patent or literature novelty. The winning configuration is a derivative plus an extended-atlas/continuation setting, and its descriptive advantage over TCFormer is uncertain. Any paper claim should be limited to the precisely implemented coordinate-field continuation and complete evaluation procedure, with equation-level prior-art comparison and ablations. Avoid "first," "unique," "reference invariant," "top globally," and "state of the art" unless separate systematic evidence supports the exact statement.